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A self-coördinating bus route to resist bus bunching

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ABSTRACT

The primary challenge for an urban bus system is to maintain constant headways between successive buses. Most bus systems try to achieve this by adherence to a schedule; but this is undermined by the tendency of headways to collapse, so that buses travel in bunches. To counter this, we propose a new method of coördinating buses. Our method abandons the idea of a schedule and even any *a priori* target headway. Under our scheme headways are dynamically *self-equalizing* and the natural headway of the system tends to emerge spontaneously. Headways also become self-correcting in that after disturbances they reëqualize without intervention by management or even awareness of the drivers.

We report on a successful implementation to control a bus route in Atlanta.

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1. Bus bunching

When buses circulate on a route, service is best when times between successive bus arrivals (headways) are equal. However, it is impossible to maintain equal headways because of variability in traffic and in the boarding and deboarding of passengers. These variations inevitably cause some buses to slow relative to others; and the larger the headway, the more strongly it tends to grow, because the trailing bus will likely meet more passengers than average at each stop and so will be further delayed. Similarly, a headway that has become smaller will tend to shrink even more because the trailing bus will likely meet fewer passengers. This phenomenon is called *bus bunching* or *platooning*.

Bus bunching increases both the mean and variance of the wait time of passengers. This is because larger gaps allow more time for passenger arrivals, and these arrivals will wait longer on average, and their waiting times will be more variable. Bunching also wastes capacity because trailing buses may be nearly empty. At the time of this writing, bus bunching is the most common customer complaint at the Chicago Transit Authority (CTA), and has received considerable local press coverage (see for example Gerasole, 2008; Luman, 2007).

We propose a method of coördinating buses that tends to equalize headways and so provides better service. In fact, under our scheme the headways become "self-equalizing" so that after any disruption the buses will tend, spontaneously, to respace themselves at equal intervals. Furthermore, this will happen without direction from management or the intention or even awareness of the drivers. Our scheme abandons the concept of a schedule and so management is freed from building and monitoring a schedule and drivers are freed from the distraction of repeatedly checking their time, location, and velocity. This seems well-suited to urban bus routes because riders typically ignore schedules with headways less than 10–12 min (Minser, 2009).

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2. Coördinating buses

Bus bunching has long been understood (for example, Newell and Potts, 1964); but it is hard to fix. There are essentially two ways to separate buses that are too close. One is to ask the leading bus to speed up; but this can be disruptive to traffic flow, and difficult or impossible in a heavily-trafficked urban environment.

The other way to separate buses that are too close is to slow the trailing bus. But a slowing bus can annoy both following traffic and on-board passengers, so in practice buses are typically delayed, when necessary, only at certain key stops, which we refer to as *control points*. The endpoints of routes that follow an out-and-back path are ideal control points because few, if any, passengers ride through them.

In the US, municipal bus routes are typically managed by target schedules, in which the arrival times of buses at each stop are planned to the minute and extra time is budgeted for each bus to pause at each control point. These delays help recover the schedule if the buses get ahead; and if behind, a bus can forfeit some or all of the planned delay. More planned delay gives greater ability to recover a schedule, but at a cost of idle bus capacity.

Recently, Daganzo (2009) proposed an approach that focuses on achieving a target headway. When a bus arrives at a control point, its headway is compared to a pre-specified target value. If the headway is smaller, then the arriving bus is judged to be following too close behind its predecessor, and it delays for a longer than nominal duration; and if the headway is larger, the arriving bus delays a shorter than nominal value.

Both target schedules and target headways attempt to realize a pre-specified static value of headway. We consider this a technical weakness in both approaches, because the ideal achievable headway is not static and not even knowable in advance. Instead, it changes continually with traffic conditions, habits of the driver, and numbers of passengers boarding and deboarding at each stop. Consequently, any system that coördinates buses based on target headways must sometimes underestimate achievable headway, and so fail to meet the target, and sometimes overestimate it, and so waste bus capacity.

A more serious objection is that neither a target schedule nor target headway can respond adequately to serious disruption. For example, when a bus breaks down, it leaves a gap until a replacement can be inserted. When the gap is large enough, it will overwhelm any planned slack. In the case of a target schedule, a trailing bus may be so far behind schedule that its planned departure time will have already lapsed when it reaches a control point, and it must depart immediately, still behind schedule. And in the case of a target headway, the delay computed for the trailing bus can be negative, which can be interpreted as directing the trailing bus to speed up—but this is not practical except in special situations, such as when there are reserved bus lanes. This leaves both schemes vulnerable to any large system-wide disruption, such as a snowstorm, that might reduce average bus velocity, and so increase all headways. When disruptions are large, both target-based schemes abdicate control, and the result can be bus bunching.

We propose a control system that abandons both the notion of a schedule and, in addition, that of any pre-specified target headway. By abandoning any target headway, the system is free to express the natural headway, which may change over time. Moreover, under our scheme headways will tend to equilibrate even in the presence of perturbations. Our system will converge to the smallest common headway possible given the current capacity and demands upon the system. Even after a large disruption of service, such as when a bus breaks down, our system will spontaneously re-position buses to achieve a new, albeit necessarily larger, common headway.

Hickman (2001) gives an excellent summary of previous work on headway control. We add the observation that the literature divides naturally into that prior to automatic vehicle location (AVL) systems, such as global positioning systems (GPS), and that after. Before GPS, approaches assumed that not much information was available. Barnett (1974) is typical in computing bus delays based on the distribution of observed headways. In these papers, the object was generally to reduce variation.

Subsequent to GPS, models have typically assumed accurate and real-time knowledge of the locations of all buses on a route, not to mention bus velocities and even instantaneous arrivals of passengers to each bus stop (Eberlein et al., 2001). Many also assume communication amongst buses (Daganzo and Pilachowski, 2011) or between buses and bus stops (Zhao et al., 2003). And, while earlier papers concentrated purely on adjusting the bus delays at control points, more recently authors have considered additional means of control, including adjusting the velocities of buses (Daganzo and Pilachowski, 2011), vehicle overtaking (Hickman, 2001), skipping some stops, or even refusing to allow some passengers to board (Delgado et al., 2009).

With the growing assumptions of more-nearly complete real-time information, authors have also been more ambitious in objective and tried to minimize some measure of the costs to passengers of waiting, both on-bus and off-bus (Hickman, 2001; Zhao et al., 2006).

We take a different approach and focus on near-term practicality. Our scheme makes use of AVL, but as little as possible, due to the practical limitations of current GPS (for example: inaccuracy during heavy rain or amongst tall buildings). Also, our scheme assumes nothing about passenger arrival rates or utilities for waiting, and therefore has no need to measure these. Rather than trying to exercise more control over bus drivers, our scheme exercises less, leaving them free to focus on driving. Our goal is simply to reduce both the mean headway and variation among headways (rather than to achieve a pre-specified target).

The control theory approach of Daganzo (2009) is closest to our work in that its data requirements are minimal and it simply corrects headways of buses departing from control points. The more recent work of Daganzo and Pilachowski

(2011) focuses, as do we, on equalization of headways, but asks buses to make coordinated adjustments of velocity in real time and relies on an estimate of passenger demand in order to define a target bus velocity (which is equivalent to defining a target headway).

In Section 3 we explain the basic mechanism of our scheme and analyze how it works in an idealized model in which perturbations are infrequent (Section 3.1). Then we enhance this model to explore the opposing tendencies of equalization and a very strong form of bunching (Section 3.2). Finally, we strengthen our scheme to produce the version that we suggest for practical use (Section 3.3).

Of course the true test of such a scheme is how well it works. In Section 4 we report on the performance of self-equalizing headways on the central bus route through the campus of the Georgia Institute of Technology in Atlanta, Georgia, US. In Section 5 we compare the three approaches—target schedule, target headway, and self-equalizing headways—by simulating ridership and traffic on route 63 of the Chicago Transit Authority. The analytic model, the field tests, and the simulations all argue for the practicality and efficacy of our scheme.

3. Self-equalizing headways

As do the target schedule or target headway approaches, our method seeks to improve service by systematically delaying buses at control points.

Consider a route with a single control point. Let the bus newly-arrived at the control point be bus 1, and index the others in the direction of travel, so that the bus trailing 1 is bus n. The headway h_i is the time separating bus i from bus $i+1$. We base the delay of bus 1 on the headway h_n of the trailing bus (or, in the terminology of some authors, on the *backwards* headway of bus 1). Specifically, we delay bus 1 at the control point for duration

$$\alpha h_n, \tag{3.1}$$

where $0 < \alpha < 1$ is a control parameter that determines the sensitivity of our scheme to perturbations.

We present three arguments for the efficacy of this scheme. These arguments are based, respectively, on an idealized model (presented in this section), on experiments with a real bus system (Section 4) and on a simulation (Section 5).

Our idealized model treats the bus route as a dynamical system with n buses moving at constant average velocity $\bar{v}$ about a circular route of length that has been normalized to 1, with a single control point at 0 (equivalently, 1). At any point in time each bus i has a location $x_i \in [0,1)$ about this circuit.

Let those instants at which a bus arrives at the control point be indexed by $t = 1,2,\ldots$. At each such time we re-index the buses so that the bus that has just arrived at the control point is bus 1, the next in the direction of bus movement is bus 2, and so on, until the last bus, which is the next bus to arrive at the control point, is bus n. For each time t let the vector $\mathbf{x}^t = (x_1^t, x_2^t, \ldots, x_n^t)$ represent the locations of the buses, where $0 = x_1^t \leqslant x_2^t \leqslant \cdots \leqslant x_n^t < 1$. From arbitrary starting positions $\mathbf{x}^0$, the trajectory of bus positions $\{\mathbf{x}^0, \mathbf{x}^1, \ldots\}$ may be thought of as a series of snapshots of the bus route at those times when a bus arrives at the control point.

Let the vector $\mathbf{h}^t = (h_1^t, h_2^t, \ldots, h_n^t)$ give the headways of the buses at time t. In the absence of perturbations, $h_i^t = (x_{i+1}^t - x_i^t)/\bar{v}$ for all buses i, except for bus 1, which we require to pause at the control point for time αh_n^t according to (3.1).

3.1. Equilibrium dynamics of self-equalizing headways

Our main result says that this method for computing delays creates a force that resists bunching. More specifically, from any initial positions of the buses—such as after a disruption of service—a common headway will spontaneously emerge. Furthermore, the value of this common headway depends only on the number and average velocity $\bar{v}$ of the buses and on the control parameter α.

Theorem 3.1. *For $0 < \alpha < 1$, any trajectory of bus positions will converge to a unique fixed point x^* with common headway*

$$h^* = \frac{1}{(n-\alpha)\bar{v}}. \tag{3.2}$$

Proof. In the absence of perturbations the positions of the buses will change from instant t to $t+1$ as described by the dynamics equations:

$$\begin{aligned} x_1^{t+1} &= 0, \\ x_2^{t+1} &= x_1^t + \left(h_n^t - \alpha h_n^t\right)\bar{v} \\ &= h_n^t(1-\alpha)\bar{v} \\ x_i^{t+1} &= x_{i-1}^t + h_n^t v \quad \text{for each } i = 3,\ldots,n, \end{aligned} \tag{3.3}$$

where the new position of bus 2 accounts for the time the bus was delayed at the control point.

Using Eq. (3.3) we can write for each $i = 3, \ldots, n$,

$$h_i^{t+1} = \frac{x_{i+1}^{t+1} - x_i^{t+1}}{\bar{v}} = \frac{x_i^t - x_{i-1}^t}{\bar{v}} + \frac{h_n^t \bar{v}}{\bar{v}} - \frac{h_n^t \bar{v}}{\bar{v}} = h_{i-1}^t \tag{3.4}$$

and so

$$h_1^{t+1} = \alpha h_n^{t+1} + \frac{x_2^{t+1}}{\bar{v}} = \alpha h_{n-1}^t + (1-\alpha) h_n^t \quad \text{(from (3.3) and (3.4))}, \tag{3.5}$$

and finally,

$$h_2^{t+1} = \frac{x_3^{t+1} - x_2^{t+1}}{\bar{v}} = \frac{x_2^t - x_1^t}{\bar{v}} + h_n^t - (1-\alpha) h_n^t = \frac{x_2^t}{\bar{v}} + \alpha h_n^t = \left(h_1^t - \alpha h_n^t\right) + \alpha h_n^t (\text{from (3.5)}) = h_1^t.$$

Thus we have the linear system

$$\begin{aligned} h_1^{t+1} &= \alpha h_{n-1}^t + (1-\alpha) h_n^t \\ h_i^{t+1} &= h_{i-1}^t \quad \text{for each } i = 2, \ldots, n, \end{aligned} \tag{3.6}$$

which can written as $\mathbf{h}^{t+1} = A\mathbf{h}^t$, where A is a stochastic matrix that may be interpreted to represent the transitions of a finite-state Markov chain that is irreducible and aperiodic if and only if $0 < \alpha < 1$. By the Markov Chain Theorem each row of the limit A^∞ is

$$\left(\frac{1}{n-\alpha}, \ldots, \frac{1}{n-\alpha}, \frac{1-\alpha}{n-\alpha}\right),$$

and all entries of $\mathbf{h}^t$ converge to the claimed common value. □

The computation of Eq. (3.6) changes the headway of each newly arrived bus to a weighted average of its former headway and the former headway of the trailing bus. If its former headway was larger, its new headway becomes smaller, and vice versa. The result is that headways are constantly adjusted to become more nearly equal. Furthermore, this adjustment proceeds from *any* starting positions of the buses, which means that the headways will tend to reëqualize after any disturbance, no matter how severe.

The smallest possible common headway for n buses, each traveling at constant average velocity $\bar{v}$ is $1/(n\bar{v})$. But such a system has no slack and so no ability to recover from disruptions. Under our scheme headways converge to the common value h^* as given in expression (3.2). From the denominator it may be seen that α represents the bus capacity held in reserve at the control point to help the system recover from disruptions.

Note that the equilibrium value h^* is determined by several factors. Slack, as given by α, is set by management. On the other hand, the value of $\bar{v}$ is mostly determined by traffic conditions and levels of ridership, which is expected to change during the day. And while management can choose the number n of buses on the route, unplanned events, such as bus breakdowns, may have an effect as well. So as conditions change, the natural equilibrium headway h^* adjusts, and under our scheme all headways will be pulled towards this common value.

It is simple to add more control points if more control of headways is wanted, such as for a long bus route or after segments with highly variable transit times. Each control point makes successive headways more nearly equal, and so Theorem 3.1 can be generalized to show that using α_j at control point $j = 1, \ldots, k$ propels convergence to a common headway of

$$h^* = \frac{1}{\left(n - \sum_{j=1}^k \alpha_j\right)\bar{v}}. \tag{3.7}$$

Others have studied the question of where to locate control points and this literature is nicely summarized by Hickman (2001). This issue seems less important under self-equalizing headways: The computations of delay at each control point are independent of those at other control points, so control points can be added or removed without interrupting operations. This can give management freedom to experiment with choice of number and location.

It is worth noting that our model can be extended, by scaling, to any common bus velocity function $v(x)$ that gives the instantaneous velocity of a bus at each point x along the route, as long as $v(x)$ is bounded above and below at every point (see Bartholdi and Eisenstein (1996), for example).

It is also straightforward to extend our analysis in a number of ways to account for sufficiently smooth and small noise as in Daganzo (2009) or Bartholdi et al. (2001). But the analysis is similar and the stability results the same.

Our idealized model demonstrates the resistance conferred by our scheme to bunching, but the real question is whether this will assert itself on an actual bus route. To answer this we first extend our analysis to allow a strong form of bunching. This will lead us to a final version of our control we suggest for implementation. We then report on both a field test and a simulation study.

3.2. Performance under strong bunching

In Section 3.1 we established the tendency of headways to equalize under our scheme. Here we show that our scheme can resist even a very strong form of bunching.

In a study of route 63 for the CTA, Milkovits (2008) measured the time a bus spends boarding and deboarding passengers at a stop. He found that passenger movement was hindered by those on the bus, and so dwell time at the stop was not simply proportional to the number of boarding/deboarding passengers, as is typically assumed, but was a super-linear function. Therefore, if the number of passengers boarding or deboarding grows linearly with the time since the last bus, the headway must grow *super-linearly*.

Such behavior may be described by letting the effective average velocity of bus i during iteration t be

$$\frac{\bar{v}}{\left(\frac{(x_{i+1}^t - x_i^t)/\bar{v}}{1/\bar{v}}\right)^{\omega}} = \frac{\bar{v}}{\left(x_{i+1}^t - x_i^t\right)^{\omega}} \tag{3.8}$$

for some $\omega \geqslant 0$. The expression $(x_{i+1} - x_i)$ measures the proportion of nominal headway for bus i, and ω imposes a super-linear decline in the velocity of the bus as the number of passengers increase. This means the velocity of a bus increases rapidly as its headway decreases.

In our enlarged model there are two opposing forces: Under the natural dynamics of super-linear dwell times, small headways tend to become smaller and large headways larger when $0 < \omega$. Contrariwise, under our system small headways tend to grow larger and large headways to become smaller when $0 < \alpha < 1$. This contention is described by incorporating (3.8) into the dynamics (3.6).

$$\begin{aligned}
x_1^{t+1} &= 0, \\
x_2^{t+1} &= \min\left\{x_1^t + \frac{T^t(1-\alpha)\bar{v}}{\left(x_2^t - x_1^t\right)^{\omega}}, x_3^{t+1}\right\}, \\
x_i^{t+1} &= \min\left\{x_{i-1}^t + \frac{T^t\bar{v}}{\left(x_i^t - x_{i-1}^t\right)^{\omega}}, x_{i+1}^{t+1}\right\}, \quad \text{for each } i = 3, \ldots, n-1, \\
x_n^{t+1} &= \min\left\{x_{n-1}^t + \frac{T^t\bar{v}}{\left(x_n^t - x_{n-1}^t\right)^{\omega}}, 1\right\},
\end{aligned} \tag{3.9}$$

where

$$T^t = \frac{\left(1 - x_n^t\right)}{\bar{v}/\left(1 - x_n^t\right)^{\omega}} = \frac{\left(1 - x_n^t\right)^{1+\omega}}{\bar{v}}$$

is the time required for iteration t, and the min operator reflects the fact that the lead bus must always be slowed (even if passing is allowed).

The dynamics Eq. (3.9) are complex, but can be examined numerically to show how the opposing forces of equalization (α) and bunching (ω) interact. Fig. 1 is typical in showing a clear boundary separating the regions of bunching and equalization. It

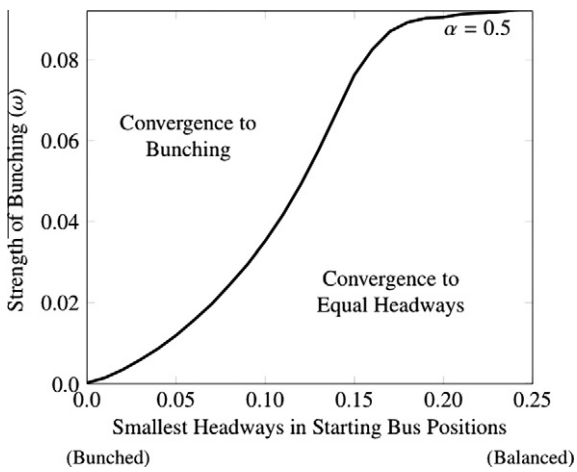

Fig. 1. There is a clear boundary separating eventual bunching from self-equalization. When the buses start with nearly equal headways (right side of the graph), then equalization overpowers the tendency to bunch. When initial headways are more out of balance, super-linear bunching can overpower our scheme (left side of the graph).

represents the limiting behavior of the dynamics Eq. (3.9) for different starting positions of four buses (the horizontal axis) versus different strengths ω of bunching (the vertical axis). The x-axis gives the starting position x_2^0 of the second bus. The third and fourth are equally spaced, so that the buses start in positions $(0, x_2^0, 2x_2^0, 3x_2^0)$. For values of x_2^0 close to 0.25, the buses begin with headways that are nearly equal, and the system tends to stay balanced. If buses begin more bunched they will converge to equalized headways if the tendency ω to bunch is not too large, but otherwise may succumb to bunching.

3.3. Strengthening headway equalization

As shown in Fig. 1, our method resists bunching and tends towards a common value of headway—unless buses become severely bunched. To help prevent severe bunching we add the stipulation that successive buses *departing* the control point be separated by at least $\beta > 0$ time units. The effect of the β is to help restrict the system of buses to the righthand region of Fig. 1, which favors evolution of equalized headways. Our scheme works as follows:

Whenever a bus arrives at a control point it must wait
- for duration αh_n; or
- until β time units after the previous bus has departed the control point,

whichever is greater.

It is advisable to set β smaller than any value of headway expected to emerge naturally. For example, it might be set to the headway expected if there were no other traffic and no passengers. So long as β has been set to a value no greater than h^*, the headways will still self-equalize.

Corollary 3.2. *In the absence of further perturbations headways will equalize under the extended scheme as long as $\beta < h^*$.*

Proof. After n iterations all headways will be larger than β, and will remain so because, in the absence of further perturbations, the minimum headway never decreases. All subsequent dynamics are as described in the proof to Theorem 3.1. □

4. Performance on a public bus route

We tested our coördination scheme on the central bus route at the Georgia Institute of Technology. This route is an out-and-back loop of total length 3.3 miles (5.3 km). It cuts through the center of campus and ties together key origins and destinations, including dormitories, the Student Athletic Center, Technology Square, and the Atlanta subway. This is the most heavily traveled of the campus bus routes with around 5000 riders each day. In addition to the usual morning and evening peaks, the route experiences surges in ridership ten minutes before and ten minutes after class changes.

To help the buses keep to schedule, the operator of the bus system (Groome Transportation) maintains two control terminals, one at each of the endpoints of the route, ISyE/Rec Center and MARTA Midtown, where the buses typically pause for a few minutes. Furthermore, a manager is stationed at the ISyE/Rec Center stop to monitor performance and deal with problems.

This route is normally run according to a schedule. Each driver is assigned a bus and each bus is assigned a repeating sequence of bus stops and corresponding times. The key performance indicator for the system and for the individual drivers was adherence to schedule.

For our experiment, we instructed the drivers to abandon the schedule and ignore headways. Instead, they were to simply drive with the flow of traffic from one end of the route to the other. A student at each control point recorded arrival times and computed departure times. This computation relies on having estimates of how long until the next bus arrives to a control point. We got this information from http://www.NextBus.com, which collects the positions of buses from their GPS devices every 15 s, predicts the number of minutes until the next bus will arrive, and updates a publicly-accessible web page.

4.1. Experiment 1: regularity of service

To be conservative we configured our scheme to approximate the normal schedule. During most of the day during the academic year there are six buses circulating the route, and the target schedule calls for 6 min headways with a 3 min wait at one of the control points and a 4 min wait at the other. The standard time allocated in the target schedule to drive the route is 29 min, and so to seek the same 6 min headway of the schedule for six buses, we wrote Eq. (3.7) as

$$6 = \frac{29}{6 - (\alpha_1 + \alpha_2)},$$

from which it follows that $\alpha_1 + \alpha_2 = 7/6$. We could have chosen any solution for which $0 < \alpha_i < 1$, but for convenience in doing hand calculations chose $\alpha_1 = \alpha_2 = 7/12 \approx 0.5$. And we set $\beta = 5$ min at each of the two control points.

As shown in Fig. 2, buses departed each control point with greater regularity under our scheme than under the target schedule. Furthermore, our headways were on average shorter than headways realized under the target schedule. Thus passenger waiting times were shorter and more reliable.

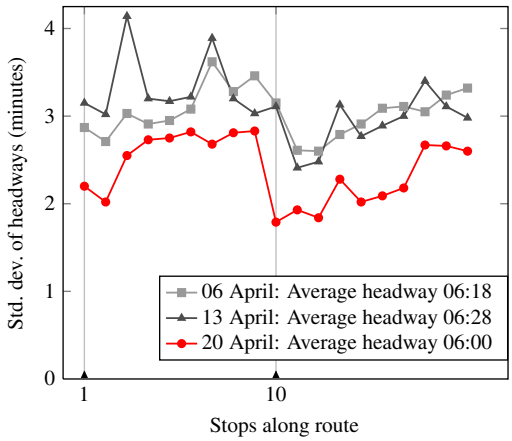

Fig. 2. Standard deviations of time between bus departures along each of the 19 stops along an out-and-back route, where stop #1 is the westernmost endpoint and stop #10 the easternmost. Under our scheme (the lowest line, in red), both the average headways and the standard deviations were reduced compared to the same day of the week in the two preceding weeks. In other words, service was both better and more reliable.

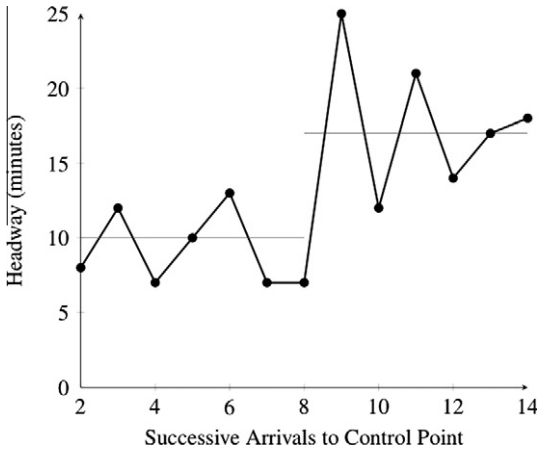

Fig. 3. Evolution of arrival headways after a bus is removed

Our scheme entirely avoided severe bunching: The smallest headway under our scheme was 2 min 28 s, compared to only 1 s under scheduled service on a comparable day. Similarly, the largest headway under our scheme was 14 min 59 s, but 19 min 45 s under scheduled service.

We interviewed bus drivers after the experiment and they liked not having to check the clock constantly, worry about the schedule, and try to make adjustments. They felt that our scheme freed them to concentrate on safe driving.

4.2. Experiment 2: resilience

In both the Georgia Tech. and the CTA systems, buses continually circle an assigned route. If a bus is suddenly and unexpectedly unavailable, such as due to mechanical breakdown, a portion of the schedule remains unserved, so that an observer at a bus stop sees a recurring gap in service. Under our scheme, the remaining buses spontaneously reposition themselves to equalize headways without any direction by management or intention of the drivers. Indeed we observed this behavior in a subsequent experiment with the Georgia Tech Bus Route. During summer term there are three buses on the route. On July 21 we removed one at an arbitrary instant and observed the subsequent headways.

The graph of Fig. 3 shows the results. The horizontal axis counts arrivals and the vertical axis the headways at the ISyE/Rec Center control point. According to the schedule, a complete loop of the route should take 36 min. But under our scheme, with three buses, the observed headway was about 10 min (significantly less than the scheduled 36/3 = 12 min). We removed one of the three buses just after arrival #8, which left a large gap of just over 12 + 12 = 24 min. Under a schedule such a large headway would be expected to grow; but under our scheme, as predicted by our model, the headways of successive arrivals reëqualized at about 17 min (slightly less than the expected 18 min).

5. Comparative performance on a simulated route

We built a simulation of route 63 of the Chicago Transit Authority (CTA). This route travels out and back along 63rd street between Stony Island Avenue and Midway Airport. The entire loop is 17.75 miles long (28.57 km). We based our simulation on CTA data collected from GPS systems and automatic passenger counters on each bus.

Route 63 has almost 80 stops, of which the CTA monitors GPS data from only 18, including the two control points, one each at the easternmost and westernmost ends of the route. The historical travel times between key stops is well-described as the sum of uniformly distributed times for each intervening city block (1/8 mile or 0.2 km in length).

We matched the simulated passenger arrivals and departures with the historical daily patterns by proceeding as follows: From the data we set the total arrivals to the system every half hour over a 14 h period, from 04:00 to 18:00. Arrivals and departures at key bus stops vary over the day according to four major time periods: AM Early, AM Peak, Midday, and Evening Peak (Fig. 4). The mean arrival rate for each particular bus stop during a given time period was estimated by sampling from an exponential distribution with mean set to the mean number of boardings observed during that period. Dwell times at bus stops were computed based on the model of Milkovits (2008).

We simulated the CTA route for a day under normal conditions and selected the best performing parameters under each of three control schemes: the target schedule used by the CTA, the target headway of Daganzo (2009), and the self-equalizing headways of Eq. (3.1).

We calibrated our simulation to best achieve the advertised headway of 7.0 min and 22 buses for the CTA schedule by setting a scheduled time of 75.0 min for the eastbound leg and 65.0 min for the westbound leg. For our self-equalizing headway control we used $\beta = 6.0$ min, one minute less than the target schedule headway. We then searched over values of α and selected $\alpha = 0.55$ as a best performer. We used the distance to the trailing bus and its average velocity to compute the h_n headways for our delay control. We used four control points, one each at the easternmost and westernmost endpoint; and one each at the eastbound and westbound crossing of the bus route with the elevated train (these provide natural control points since so many passengers board and deboard at these stops).

Under the target headway scheme of Daganzo (2009), a bus arriving at the control point is delayed for duration

$$\max\{0, d + g(H - h_1)\}, \tag{5.1}$$

where H is the target headway, h_1 is the *forward* headway of the arriving bus, $d > 0$ is the average delay at equilibrium, and $g > 0$ is a control parameter. For targeted headway control we simulated over a thousand parameter combinations, including the selection of either four or eight control points, and chose the best performing: $d = 5.8$, $g = 0.8$, $h = 7.0$, and eight control points.

When each scheme is allowed to choose its best parameters *ex post facto*, self-equalizing headways performed best, with an average wait time over the day of 3.92 min, compared to 4.79 min for the target schedule, and 6.15 min for the target headway method.

Of course, in practice one cannot select parameters *ex post facto*. The control scheme must be able to react to the shocks and variances of each day. To test this, we used the parameter settings described above, but re-ran the simulation to mimic a reduction in travel velocity by 10%, as might be caused by bad weather or road construction.

The results are shown by the solid lines in Fig. 5. Initially, all schemes performed well under the stress of decreased bus velocity. The target schedule was the first to succumb to bunching (indicated by the increased standard deviation of bus headways). The target schedule was able to recover during the midday lull, but at 14:00 both the target schedule and target headway allowed a rapid increase in bunching. Furthermore, neither recovered before the end of the day. In contrast, self-

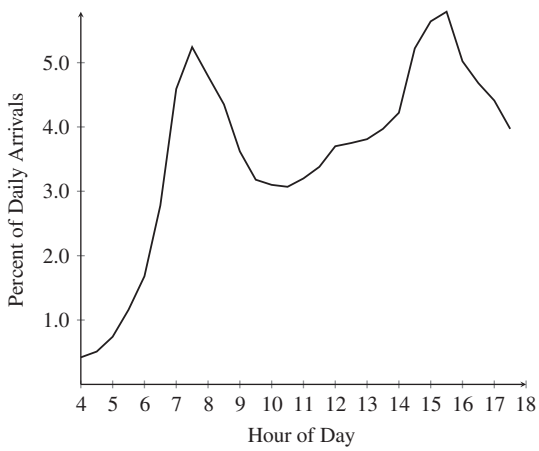

Fig. 4. Arrival rates of passengers to CTA route 63, showing morning and evening surges

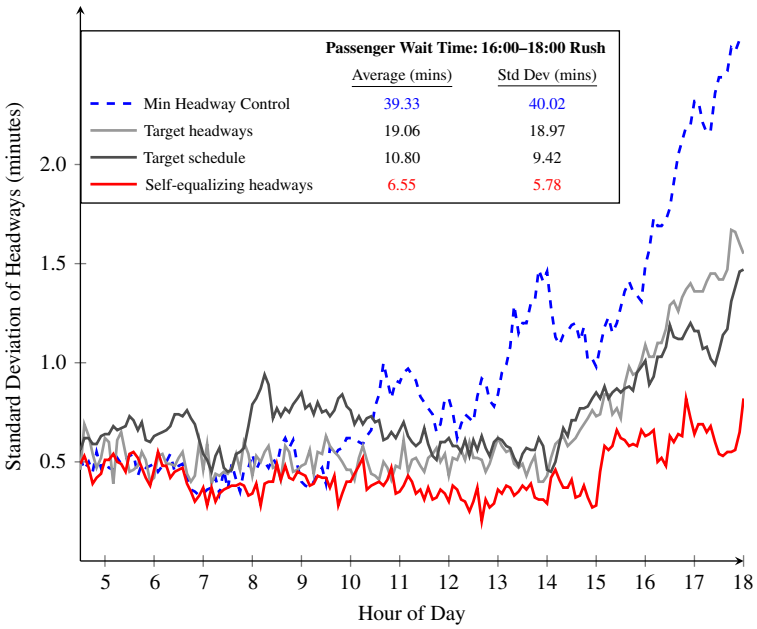

Fig. 5. Standard deviation of bus headways over a simulated day of heavy ridership.

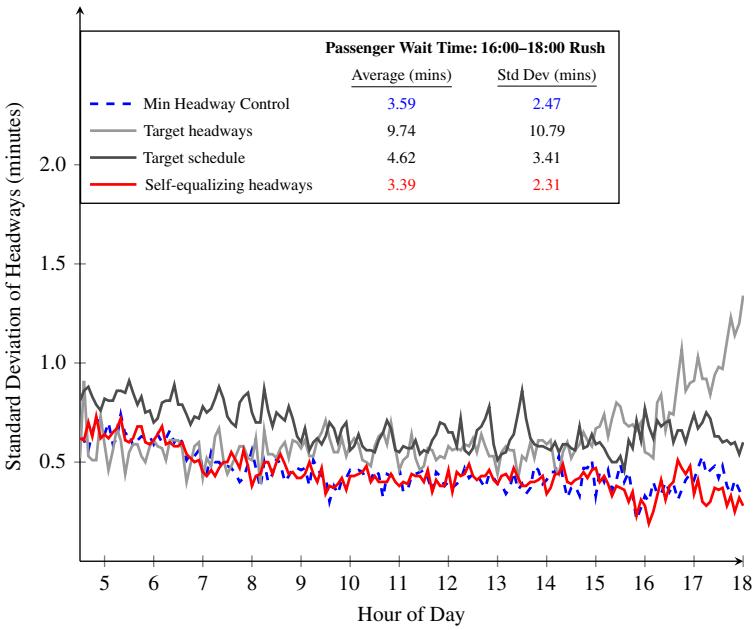

Fig. 6. Standard deviation of bus headways over a simulated day of light ridership.

equalizing headways showed less degradation during surges in ridership and recovered more quickly. The table within Fig. 5 shows the resulting average and standard deviation of passenger wait times during the early evening peak. Under self-equalizing headways passengers had significantly shorter waits on average and much less uncertainty.

Part of the reason that neither target schedules nor target headways worked well is that neither is able to adapt sufficiently to the reduction in system capacity when traffic slowed. We believe this is because they rely too much on an accurate forecast of achievable headway, which is in turn based on achievable velocity. But velocity depends on many factors outside the control of management. In contrast, our method does not attempt to guess achievable headways or velocities, and is independent of them. (The parameter β is a loose lower bound, that will not interfere if the resultant headway increases.)

Fig. 5 also shows (with the blue dashed line) a simple minimum headway control which is simulated by setting $\alpha = 0$ and keeping $\beta = 6.0$. This demonstrates that the results of our algorithm are not due to simple minimum headway control, but that the self-equalizing influence of our α parameter is critical to performance.

We also ran the CTA simulation assuming a lightly loaded day (an increase of 10% in bus velocity and a decrease of 1/6 in average passenger arrivals). Fig. 6 shows the results. All methods perform better, as to be expected; but self-equalizing headways and the minimum headway control perform best, because both are able to take advantage of the fact that the achievable headway has decreased.

6. Conclusions

Under our control scheme the system of buses acts in effect as an analog computer and, in the absence of perturbations, computes the common headway by achieving it. This reduces work for management, and simplifies the job of the drivers, who can focus purely on flowing with the traffic. And most importantly, it provides better service to riders.

It is unnecessary for management to construct a schedule in advance or to monitor adherence to a schedule. All that is required is the setting of two parameters, α to determine bus utilization and speed of convergence of headways; and β, a lower bound on headway to strengthen equalization. The amount α of bus capacity held in reserve is easy to understand and to control. A manager can adjust the resultant headway by changing the control parameter α or by inserting an additional bus into the flow or by removing one, and the headways will reëquilibrate appropriately.

From (3.6) equalization of headways is slowed for α near 0 or near 1. Analysis for small number n of buses and computational experiments for larger n suggest that convergence (self-equalization) is fastest for values of α in the range 0.5–0.6. Using an α near 0.5 has, so far, always been accepted by managers of bus systems, as this choice means that at equilibrium each bus is delayed by half the natural headway; about 3–5 min in the small headway systems for which our approach is best suited. In any case, the Markov chain of Theorem 3.1 converges exponentially fast and therefore the time for our system to recover equal headways after a disturbance grows only logarithmically in the size of the disturbance.

Our scheme responds even to large disruptions, such as a bus breakdown or surge in ridership, by re-distributing the buses to equalize the headways at a new (larger) value. Similarly, with reduced traffic, the common headway is spontaneously reduced and service improves.

Another advantage of our scheme is that buses can be added to or removed from a route at arbitrary times and points. This is not so for a target schedule: For example, on the Georgia Tech. Bus Route, scheduled service starts early in the day with a single bus, then increases to three, and finally to six; and in the evening service is reduced from six to three to one. It is impractical to increase the number of buses from three to, say, four, because the schedules of at least two of the three buses would have to be re-anchored. In contrast, under our scheme, buses can be added or removed arbitrarily (though one may wish to adjust β and perhaps α).

Our method is also scalable, in the sense that one can further reduce variance in headways simply by adding more control points, which enables more frequent correction.

Our scheme is robust even when the route is reconfigured, such as when construction altered the flow of traffic on campus and forced the temporary re-routing of the Georgia Tech. buses. Such changes would require construction of a new target schedule; but our scheme would continue to produce regular bus service.

Under our scheme any improvements in processes, such as introducing procedures to speed boarding, go directly to reduce headways, without the need to rebuild schedules or re-compute target headways.

Our scheme is easy to implement. In the first experiment students computed the departure times for each successive bus. For the second experiment we provided the same functionality in wifi-enabled netbook computers. We are now building the third generation of the control system on mobile phones to be mounted in each bus. A subsequent paper will describe details.

Because of its simplicity it is easy to adapt our scheme to account for additional business rules. For example, some bus systems guarantee drivers a short break time,γ, at each arrival to specified control points. In this case, our scheme can be extended to delay each bus for either $\gamma + \alpha h_n$ time units or until the previous bus has been gone for β time units, whichever is later. Self-equalization continues to assert itself: The argument of Theorem 3.1 holds, but with the time $1/\bar{\nu}$ to drive the route extended by the required break time γ.

We think this system may be useful for airport shuttles, such as for parking or rental cars. It may also be useful when a transit system sets up a temporary service for a large public event, such as a state fair or sporting event. In such cases there may not be enough history of traffic velocities or ridership to produce a believable schedule.

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